

EXPERIMENT- 04

Python Code:

```
04.py > ...
1  plaintext = input("Enter plaintext: ")
2  key = input("Enter key: ")
3
4  extended_key = key * (len(plaintext) // len(key)) + key[:len(plaintext) % len(key)]
5
6  encrypted_text = ""
7  for i in range(len(plaintext)):
8      if plaintext[i].isalpha():
9          shift = ord(extended_key[i].lower()) - 97
10         if plaintext[i].isupper():
11             encrypted_text += chr((ord(plaintext[i]) - 65 + shift) % 26 + 65)
12         else:
13             encrypted_text += chr((ord(plaintext[i]) - 97 + shift) % 26 + 97)
14     else:
15         encrypted_text += plaintext[i]
16
17  print("Encrypted Text:", encrypted_text)
18
19  decrypted_text = ""
20  for i in range(len(encrypted_text)):
21      if encrypted_text[i].isalpha():
22          shift = ord(extended_key[i].lower()) - 97
23          if encrypted_text[i].isupper():
24              decrypted_text += chr((ord(encrypted_text[i]) - 65 - shift) % 26 + 65)
25          else:
26              decrypted_text += chr((ord(encrypted_text[i]) - 97 - shift) % 26 + 97)
27      else:
28          decrypted_text += encrypted_text[i]
29
30  print("Decrypted Text:", decrypted_text)
```

Output:

```
PS D:\College\Crypto\Experiments> python -u "d:\College\Crypto\Experiments\04.py"
Enter plaintext: hello world
Enter key: key
Encrypted Text: rijvs gspvh
Decrypted Text: hello world
PS D:\College\Crypto\Experiments> |
```